1. **Aim of the practical:** To understand and demonstrate managing SE Linux Security

2. **Objective:** The objective is to understand and manage SELinux effectively by checking its current status, modifying its operational mode (Enforcing, Permissive, or Disabled) within the configuration file, and applying the changes through a system reboot, ensuring proper security policies are enforced based on the desired mode.

3. Hardware and Software Requirements:

a) **Disk Drive**: Physical disk (HDD/SSD) with enough space.
   Computer System: Host system running virtualization software.
b) **Operating System**: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: Check SELinux status: Verify current SELinux mode.

```
nikhilyadav@fedora:~$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
```

Step-2: **Open config file**: Access SELinux settings file.

```
nikhilyadav@fedora:~$ sudo vim /etc/selinux/config
[sudo] password for nikhilyadav:
```

Step-3: **Locate SELINUX= line**: Find mode setting to change.

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
                                                        25,1          Bot
```

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=permissive
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
-- INSERT --                                          22,19         Bot
```

Step-4: **Save and exit**: Store changes in the config.

```
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#    grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#    grubby --update-kernel ALL --remove-args selinux
#
SELINUX=permissive
# SELINUXTYPE= can take one of these three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are pro
tected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted


~
:wq!
```

Step-5: Verify current SELinux mode.

```
ikhilyadav@fedora:~$ sestatus
ELinux status:                 enabled
ELinuxfs mount:                /sys/fs/selinux
ELinux root directory:         /etc/selinux
oaded policy name:             targeted
urrent mode:                   permissive
ode from config file:          enforcing
olicy MLS status:              enabled
olicy deny_unknown status:     allowed
emory protection checking:     actual (secure)
ax kernel policy version:      33
ikhilyadav@fedora:~$
```

6. Learning outcomes (What I have learned):

1. SELinux has three modes: Enforcing (actively enforces security policies), Permissive (logs violations without enforcing), and Disabled (turns off SELinux).
2. SELinux configuration is managed in /etc/selinux/config, where you can set the desired mode by modifying the SELINUX= line.
3. Changes to SELinux mode require a system reboot to take effect.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature